Predictive Analysis Final Examination for the Class of 2023 (Recall Version)

Problem 1. Suppose that the true data-generating model is

$$y = h(x) + \varepsilon,$$

where $\mathbb{E}(\varepsilon) = 0$, $\mathrm{Var}(\varepsilon) = \sigma^2$, and the noise ε is independent of the training set D. Let $f(x, D)$ denote the prediction function obtained from the training set D. For a fixed input x, answer the following questions:

(1) Under squared loss, fully derive the bias–variance–noise decomposition of the expected prediction error

$$\mathbb{E}_{D,\varepsilon}\Big[(f(x, D) - y)^2\Big],$$

and clearly state the meaning of each term.

(2) Briefly explain the general relationship between model complexity, bias, and variance. Also state common methods for choosing an appropriate model complexity.

Problem 2. Let $Y(x) \sim \mathcal{GP}(0, C(x, x'))$ be a one-dimensional Gaussian process with covariance function

$$C(x, x') = 4\exp\Big[-(x - x')^2\Big].$$

Suppose that the observations are $Y(0) = 2$ and $Y(2) = -2$. Answer the following questions:

(1) In the noiseless observation case, find the posterior predictive distribution of the latent function value $Y(1)$ given the observations.

(2) Suppose instead that the observation model is

$$t(x) = Y(x) + \eta, \qquad \eta \sim \mathcal{N}(0, \tau^2),$$

where the observation noises are mutually independent and independent of the Gaussian process. Given $t(0) = 2$ and $t(2) = -2$, find the posterior predictive distribution of the latent function value $Y(1)$. If the prediction target is the noisy observation $t(1)$, also give the corresponding predictive distribution.

Problem 3. Given a linearly separable training set $\{(x_n, y_n)\}_{n=1}^{N}$, where $x_n \in \mathbb{R}^d$ and $y_n \in \{-1, +1\}$, consider the hard-margin support vector machine primal problem

$$\min_{w,b}\ \frac{1}{2}\|w\|^2, \qquad \text{subject to } y_n(w^\top x_n + b) \geq 1, \quad n = 1, \ldots, N.$$

Answer the following questions:

(1) Write down the Lagrangian and the KKT conditions for this optimization problem, and derive the corresponding dual optimization problem.

(2) Using the KKT conditions, explain why the data points that truly determine the decision boundary usually account for only a small subset of the training data.

Problem 4. Let $X = \{x_n\}_{n=1}^N$ consist of independent observations. For each n,

$$x_n = (x_{n1}, \dots, x_{nD})^\top \in \{0,1\}^D.$$

The data are generated from a two-component mixture of multivariate Bernoulli distributions. The class priors are $\phi_1 = \phi$ and $\phi_2 = 1 - \phi$. Conditional on class $k \in \{1,2\}$, the coordinates are conditionally independent and

$$p(x_n \mid z_{nk} = 1) = \prod_{d=1}^{D} \mu_{kd}^{x_{nd}} (1 - \mu_{kd})^{1-x_{nd}}.$$

Introduce a one-hot latent variable $z_n = (z_{n1}, z_{n2})$, where $z_{nk} \in \{0,1\}$ and $\sum_{k=1}^2 z_{nk} = 1$. Answer the following questions:

(1) Write down the complete-data log-likelihood in a form convenient for optimization.

(2) Derive the EM algorithm for this mixture model. Give the full derivation of the E-step and the M-step, and write down the parameter update formulas for ϕ and μ_{kd}.

Problem 5. Let x be an observed variable, z be a latent variable, $p(x,z)$ be the joint distribution, and $p(z \mid x)$ be the posterior distribution. Answer the following questions:

(1) For an arbitrary variational distribution $q(z)$, derive

$$\log p(x) = \mathcal{L}(q) + \mathrm{KL}\Big(q(z) \,\|\, p(z \mid x)\Big),$$

and write down the explicit forms of the evidence lower bound $\mathcal{L}(q)$ and the KL-divergence term.

(2) Suppose that the variational distribution has the mean-field factorization

$$q(z) = \prod_{j=1}^{M} q_j(z_j).$$

With all factors $q_i(z_i)$ for $i \neq j$ fixed, derive the optimal update for the j-th factor:

$$\log q_j^*(z_j) = \mathbb{E}_{q_{-j}}\Big[\log p(x,z)\Big] + \text{constant}, \qquad q_{-j}(z_{-j}) = \prod_{i \neq j} q_i(z_i).$$